LEARNER GUIDE

Letters as unknowns and variables

Formulae and Rearranging · Lesson 1. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

In algebra a letter stands for a number. In $x + 5 = 12$ it is one particular number we have not found yet: an **unknown**. In P = 2l + 2w the letters can take many values and change together: they are **variables**. There are four kinds of statement. An **expression** (4a + 2) has no equals sign: it has a value, but makes no claim. An **equation** ($2x + 3 = 11$) is true only for particular values. A **formula** (A = lw) links quantities, so one is worked out from the others. An **identity** ($3(n + 2) = 3n + 6$) is true for every value.

What you are learning to notice

- To decide what kind of statement you have, ask **when it is true**: for some values, for every value, as a link between quantities — or is there no equals sign at all?
- $3n$ means $3 \times n$. n^2 means $n \times n$. $n/2$ means $n \div 2$.
- A power belongs only to what it is attached to: $3n^2 = 3 \times n^2$, but $(3n)^2 = 3n \times 3n$.
- Write the number first ($4n$, not $n4$) and leave out the × sign.

Moving through it, one step at a time

Test a statement with several values, including 0 and a negative number. If it is true for just some of them, it is an equation. If you cannot find any value that makes it false, it is probably an identity — and one counterexample is enough to show a statement is not an identity. When you substitute, write the meaning out in full first: $3n^2$ with $n = 2$ is $3 \times 2 \times 2$.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Sorting statements by testing values

What kind of statement is each of $5y = 20$, $n + n = 2n$ and 7p – 1?

1. $5y = 20$: true for $y = 4$, false for $y = 0$.	an equation
2. $n + n = 2n$: true for $n = 3$, $n = 0$, $n = -5$ … for every value.	an identity
3. 7p – 1: no equals sign. It has a value, but makes no claim.	an expression

An equation, an identity and an expression.

Example 2 — Reading the notation when substituting

Work out $3n^2$ and $(3n)^2$ when $n = -2$.

1. $3n^2$ means $3 \times n^2$: only n is squared.	$3 \times (-2)^2 = 3 \times 4 = 12$
2. $(3n)^2$ means the whole of $3n$ is squared.	$(3 \times (-2))^2 = (-6)^2 = 36$
3. Put the negative number in brackets.	$(-2)^2 = 4$, not -4

$3n^2 = 12$ and $(3n)^2 = 36$.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (the order of operations (multiply before you add)), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.